

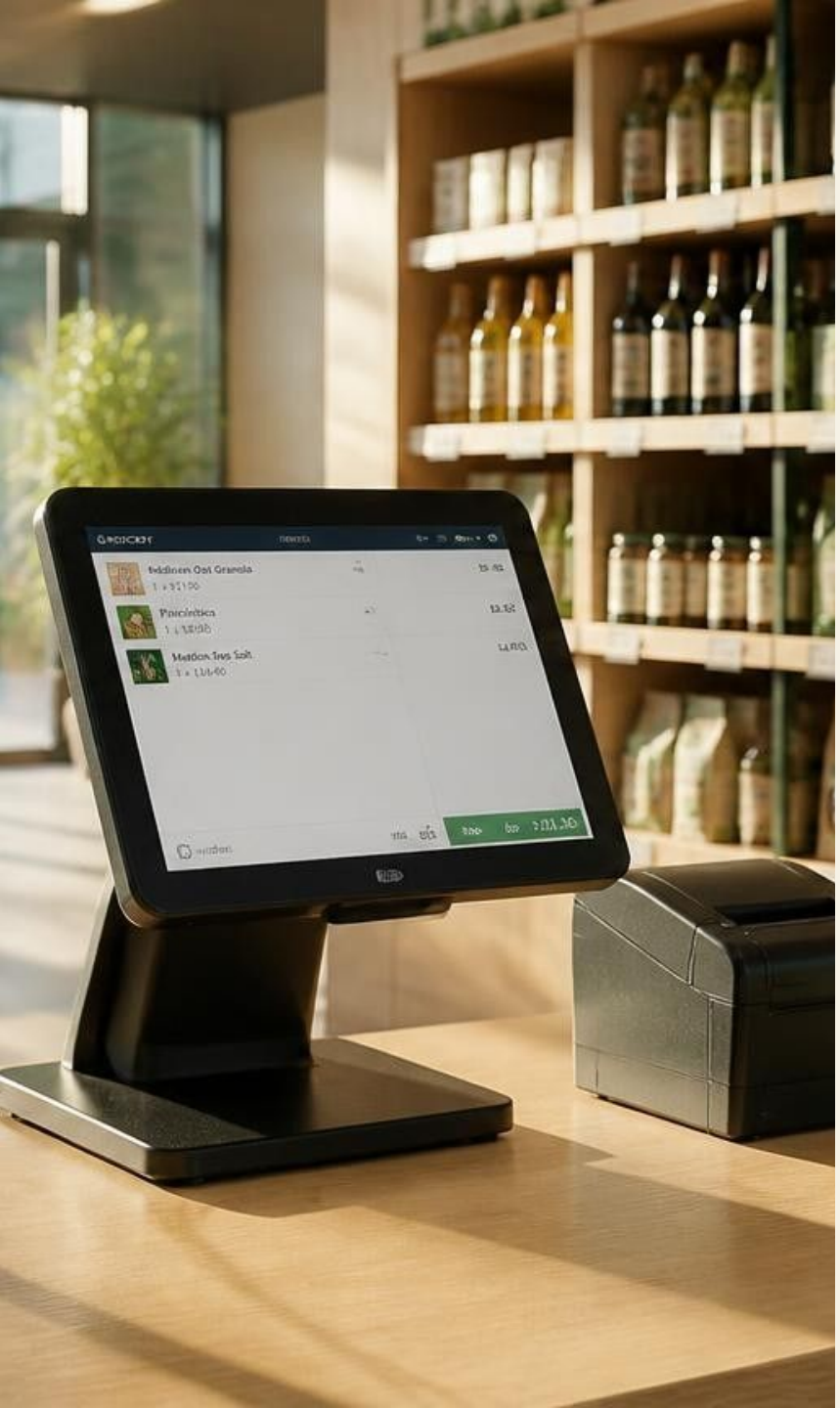
PROJECT PROPOSAL · RETAIL SYSTEMS PLANNING · 2026

Product & Inventory Management System

A smart, unified business workspace planned for independent retail shops in Bangladesh.

Scope: Core MVP system architecture · Prepared for project evaluators & business owners

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One system for the work behind every sale

Bring products, stock, transactions, credit, supplier records, cash flow, and profit visibility into one practical workspace.

- Centralized product and stock management
- Simplified sales and purchase tracking
- Organized customer credit and supplier records
- Clearer cash-flow and profitability visibility

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OPERATING VULNERABILITIES

Manual records create everyday business risk

Paper ledgers, informal tally sheets, and memory introduce silent leakage at every operating stage.



Stock counts drift from reality

Unrecorded sales, misplaced cartons, and unaccounted damages lead to unexpected stockouts and missed revenue.



Selling prices fall out of date

When wholesale purchase costs rise, staff sell at outdated sticker prices, quietly eroding gross profit margins.



Drawer discrepancies hide true earnings

Cash withdrawals for petty expenses and personal draws blend into trading cash, obscuring real operating profit.



Expired goods become easy to miss

Perishable staples sit unnoticed behind newer stock, resulting in unavoidable inventory write-offs and direct capital loss.



Customer credit records get scattered

Informal customer credit accounts (Baki khata) written on loose sheets lead to missed repayments and customer disputes.



Supplier payables are hard to reconcile

Without structured purchase histories, shopkeepers risk duplicate vendor payments or uncollected return credits.

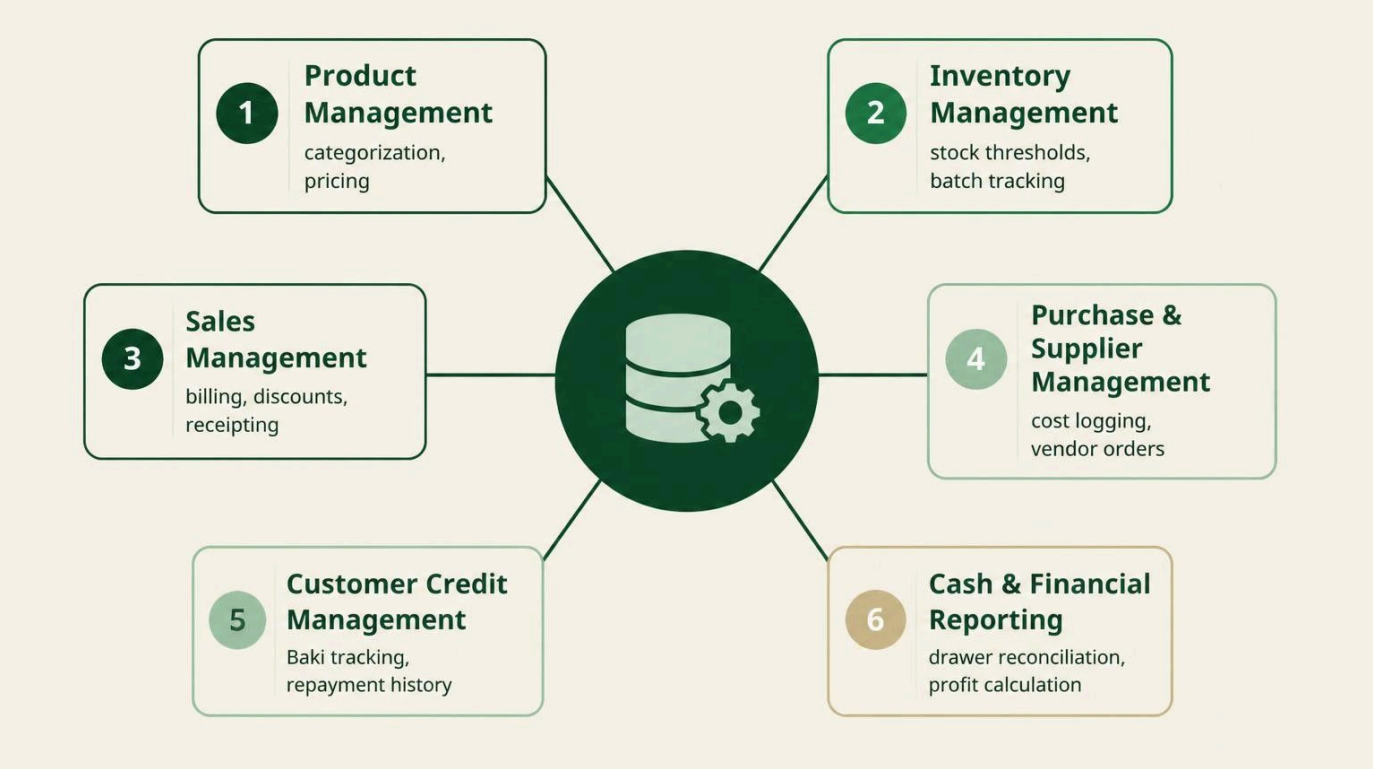
Problem analysis: Field observations of small retail trading habits and informal bookkeeping practices in suburban markets.

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MODULAR ARCHITECTURE

A connected platform for the whole shop

Six specialized operational modules interact through a shared database core, keeping records synchronized in real time.



System architecture: Planned modular software blueprint ensuring atomic transaction isolation across retail operations.

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INVENTORY CONTROL

Know what is in stock, what it costs, and what needs attention

Stock visibility begins with complete catalog control. The system records dual pricing tiers, enforces minimum replenishment thresholds, and archives every stock adjustment with batch provenance.

- **Dynamic cataloging:** Categorize items, record purchase prices, and update retail selling rates instantly.
- **Automated threshold alerts:** Flag low-stock items before shelves run empty to protect daily revenue.
- **Batch & expiry tracking:** Ensure first-in, first-out stock movement to eliminate product spoilage.
- **Movement logs:** Maintain comprehensive audit trails of every incoming shipment and customer sale.

Data model: Real-time inventory state tracking with threshold-triggered replenishment flags.

Illustrative product inventory monitor

PRODUCT	QTY	PURCHASE	SELLING	STATUS
Rice 5 kg	24	৳420	৳480	Healthy
Milk powder	6	৳310	৳360	**Low stock**
Biscuits	18	৳55	৳70	Healthy
Cooking oil 1 L	3	৳165	৳190	**Low stock**

51 units

MONITORED SAMPLE STOCK

2 items

TRIGGERING REORDER ALERTS

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OPERATING CYCLE

Every transaction updates the next one

Inventory, payment ledgers, and profit calculations stay connected through a synchronized workflow.

01

1. Procurement

Log supplier invoice, purchase price, and item quantity.

02

2. Inbound restock

Available stock increases instantly with batch cost assigned.

03

3. Point of sale

Scan products, calculate discounts, and select cash or credit payment.

04

4. Stock reduction

System deducts inventory and commits revenue to the drawer.

EXCEPTION BRANCH · RETURNS & ADJUSTMENTS

Customer returns automatically restore sellable stock quantities and adjust the day's gross sales receipts, while supplier returns reduce outstanding accounts payable without manual reconciliations.

Zero ledger drift guarantee

Transaction lifecycle: Standardized state machine tracking items from vendor intake to customer checkout.

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LEDGER ACCOUNTABILITY

Credit and supplier records stay accountable

Replace untracked memory and scattered notebooks with structured balances, repayment histories, and proactive vendor schedules.

Customer credit (Baki)

RECEIVABLES

Receivables & repayment ledger

- **Verified contact profiles:** Store names, phone numbers, and addresses.
- **Credit sale logs:** Itemize items taken on credit with timestamps.
- **Repayment records:** Log partial and full cash settlements cleanly.
- **Overdue monitoring:** Generate balance reminders to speed collection.

Supplier payables

PAYABLES

Procurement & vendor accounts

- **Vendor directories:** Maintain distributor terms and contact cards.
- **Invoice matching:** Connect received stock directly to unpaid bills.
- **Settlement history:** Track partial payments, bank slips, and cash paid.
- **Return adjustments:** Deduct returned or damaged cartons from dues.

Control objective: Eliminating disputed customer debt while building trustworthy supplier credit history.

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FINANCIAL DISTINCTION

Cash balance and profit answer different questions

A full cash drawer does not always mean a profitable month, and strong sales revenue can coexist with cash deficits if customers buy on credit or supplier bills come due.

Separate cash movement from profit calculation, then reconcile expected and actual closing cash.

Dual reconciliation

CASH DRAWER VERSUS INCOME STATEMENT

Reconciliation: Compare expected cash with actual closing cash.

Path 1: Physical cash movement

Tracks money physically inside the shop drawer

Opening cash + Sales receipts + Repayments − Supplier payments − Expenses − Owner withdrawals = **Expected closing cash**

Path 2: Profit & loss calculation

Tracks trading performance independent of cash timing

Sales revenue: Completed sales, adjusted for returns and discounts

Gross profit: Net sales − cost of goods sold

Estimated net profit: Gross profit − operating and other applicable expenses

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DECISION INTELLIGENCE

A dashboard turns records into daily decisions

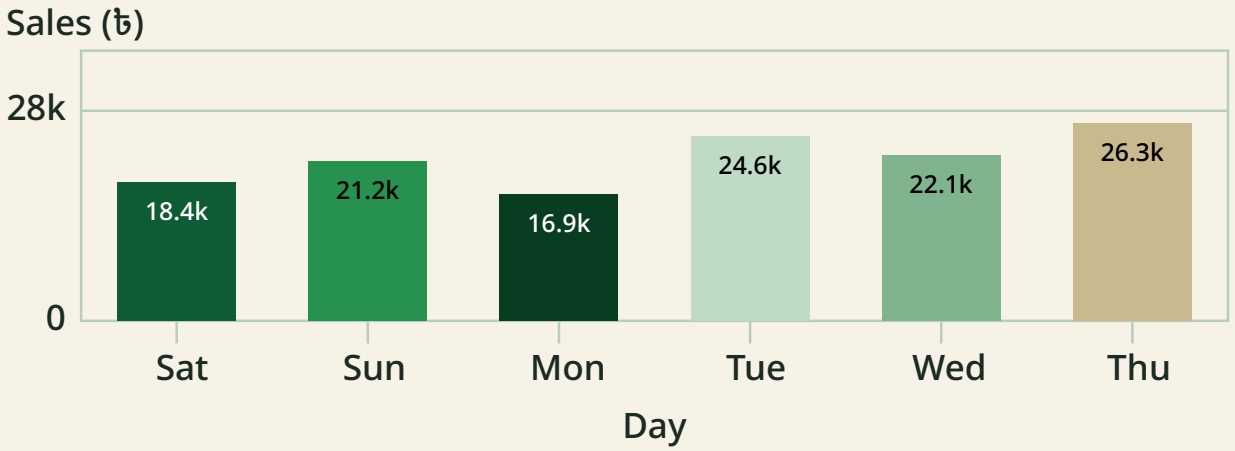
Illustrative daily sales trend (sample data only · Saturday to Thursday)

৳119,500

Weekly sales volume (illustrative sample)

৳19,917

Average daily trading run rate



Real-time executive cards

- Today's sales and gross profit
- Estimated net profit and inventory value
- Low-stock and approaching-expiry alerts
- Outstanding customer credit and supplier payables
- Daily cash summary and best-selling products

Illustrative notice: Data shown illustrates reporting visualization capabilities and does not reflect actual store performance claims.

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Simple for the shop floor, dependable behind the scenes

User experience

Mobile-friendly transaction screen

Fast product search

Clear forms · Validation messages

Mobile-friendly interface, simple navigation, and Bangla-language support.

Security & reliability

Clear roles. Dependable records.

Owner

Employee

Secure authentication · Role-based permissions · Audit logs · Database backups

Reliable operations

Transaction-safe inventory updates plus recovery planning.

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